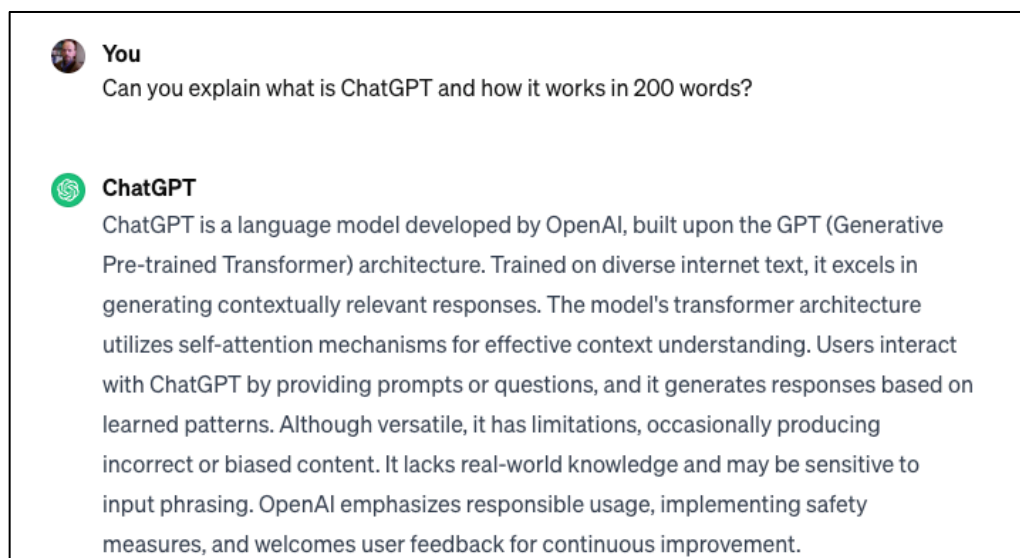


Chapter 1: How do large language models work, and what are they a model of?

José-Luis Mendívil-Giró

If we ask ChatGPT (the most popular of the latest generation of large language models) what it is and how it works, it writes—in a couple of seconds—the text we have in figure 1.1.

Figure 1.1: ChatGPT describes itself



Although the reader will have understood the definition—which is essentially correct—the wonderful thing is that the algorithm that wrote it does not understand a word of what it says. It also didn't know that humans would be reading its text, and it doesn't even know what a human is.

Nevertheless, it is clear that ChatGPT is an impressive and sophisticated engineering tool, as well as extraordinarily useful. Its potential, including its commercial use, is set to be enormous. Without a doubt, for better and worse, in natural language processing technology there will be a before and after marked by this latest generation of large language models (LLMs). But surely there will be no such radical change, contrary to what numerous authors have suggested, with regard to our knowledge of human language and the human mind.

1. The human brain and the Chinese room

The Chinese Room is a renowned thought experiment proposed by John Searle to challenge the notion that machines are capable of understanding and processing information in a manner analogous to that of humans. In the *Gedankenexperiment*, Searle postulates a scenario in which an individual who is not proficient in the Chinese language

(Searle himself) is confined within a room. The individual has a set of rules that enables him to manipulate Chinese symbols in response to specific inputs. To illustrate, if presented with a question in Chinese, the imaginary Searle will consult the rules and produce a response in Chinese, despite lacking a genuine comprehension of the language. Searle posits that, although the individual may produce responses that appear to be appropriate to native Chinese speakers, he does not actually comprehend the language. Similarly, Searle posits that a machine that emulates this form of symbol processing cannot possess a genuine comprehension of language or the world in the same manner as humans.

Of course, some philosophers have argued that Searle's thought experiment does not serve to reject a real knowledge capacity of Artificial Intelligence (AI), since it may well be that our own brains (the standard we use to define *knowledge*) are also Chinese rooms. We should then conclude that even if there were a real Chinese speaker inside the famous room, her brain would function like the Chinese room she is in. In fact, there is no doubt that, at a certain level, our "knowledge," whether of Searle, of Chinese, or of ourselves, will be supported by groups of cells and tissues that behave like the individual in the Chinese room, or like the algorithm that powers ChatGPT: moving chunks of incomprehensible information from one site to another. Nevertheless, even if this is the case, it does not necessarily follow that we cannot assert that we possess knowledge and that we are aware of our possession of that knowledge.

In any case, beyond the deep paradoxes and mysteries of consciousness and the very nature of knowledge, the relevant question now is whether LLMs such as ChatGPT have a knowledge of language that is analogous or comparable to that which human beings have. The many scientists who believe so are those who defend that ChatGPT is a useful model to better explain how it is possible that we can learn and use the languages of the environment—and even some, like Steven Piantadosi (2023), affirm that these systems are authentic theories of language, while scientists who believe not—for example, Chomsky, Roberts and Watumull (2023)—are those who reject that LLMs are relevant to better understanding our language capacity, even though they are useful as engineering products.

In order to understand the manner in which these inconsistencies arise, it is essential to initially examine the fundamental mechanisms underlying the operation of these LLMs. It is only then that an evaluation can be made as to whether the knowledge currently held about the processes of language acquisition and use in the human brain is comparable or equivalent to that observed in these AI models.

2. Inside the stochastic parrot: how does ChatGPT work?

Computational linguist Emily Bender has popularized the name *stochastic parrots* for LLMs as it encapsulates two pivotal characteristics of their operation: the reliance on probability-based word prediction and the absence of comprehension.

And, in fact, what LLMs really do is calculate the probability that a graphic word appears taking into account the graphic words that it already has chained together.¹

¹ In this brief description of how current LLMs work, I focus on ChatGPT and I rely on the (much more detailed and technical) characterization by Stephen Wolfram (2013).

I have purposely used the expression “graphic word” because, unlike what happens with human language, ChatGPT does not operate with authentic linguistic units (that is, morphemes and lexemes, entities with meaning), but with tokens, that is, sequences of very frequent combinations of letters or characters. Computational linguists had previously identified that the outcomes are enhanced when these tokens, which comprise approximately four characters in English, are utilized. Consequently, a single token is deemed to be analogous to approximately “three-quarters” of a graphic word in English (100 tokens = 75 graphic words). The tokens that ChatGPT manipulates are, in some ways, the written language equivalent of syllables in oral language: recurring sequences of basic units with no correlation to meaning. There is, apparently, no theoretical motivation behind this practice, but rather greater efficiency and flexibility in the task of predicting the next *word* (hereinafter understood as “frequent sequence of characters”) taking into account practically everything that has been written in English and was digitized.²

The key question is how to obtain the probabilities that the program uses to evaluate which word to write next to the one you just typed. Note that it is not about calculating the probability of the next word taking into account the last one used, but taking into account those already used (otherwise, coherent and natural-looking text would not be generated, but rather a confusing and repetitive sequence).

However, Wolfram’s calculations indicate that the quantity of written English text in existence is insufficient to enable the calculation of the probabilities associated with groups of words. Given a lexicon of approximately 50,000 words, the potential combinations of three-word sequences exceed 60 trillion. For 20-word sequences (a typical length of written sentences), the number of possibilities is greater than the number of particles in the universe. Consequently, there is not—nor could there ever be—enough written text in the world for those probabilities to be calculated.

So what ChatGPT does to instantiate its basic ability to guess how to continue the piece of text it is given is to use a *model* (a huge mathematical function) that allows it to estimate the probabilities of certain sequences of words appearing after certain sequences of words, even though those sequences have never appeared in the training corpus. Thus, ChatGPT is essentially a mathematical model of the distribution of tokens in the enormous corpus with which it has been trained. It should be noted that a model is a mechanism for calculating specific values without the necessity of measuring them individually. This is exemplified by the use of a function to represent, for example, the approximate trajectory of an object (chapters 2 and 3 in this volume deal specifically with this notion of model).

To construct this model, ChatGPT consists of a huge neural network that has been trained on a large corpus of text, about 500,000,000 words, according to Wolfram. A crucial aspect of training this neural network to discern coherent sequences of text is to provide it with incomplete texts (where a fragment has been removed from the end) and utilize the complete text as the training objective output. This approach allows the network to gradually learn to complete the text, eventually producing the original text or a highly similar one. Once this is achieved, the weights or numerical values of each node in the network are set. These weights subsequently facilitate the calculation of probabilities.

Each of the 100,000 tokens used by ChatGPT is characterized by a list of numbers. Starting from there, frequent sequences of tokens can also be characterized by lists of

² For a detailed explanation of the tokenization process in ChatGPT, see Manova’s chapter in this volume.

numbers. And that's what ChatGPT does when it interacts with the user: it takes the text that is given to it (the prompt that the user enters) and converts it into a vector of numbers that it processes. Since there are about 100,000 tokens in the dictionary, what you get is a list of 100,000 numbers giving the probabilities of each of the possible words. In a very clever way, ChatGPT does not always use the most probable word, but randomly chooses from time to time one with a lower probability (from the top of the list). Apparently, this has the effect of making the texts less “flat” and more original. This “freedom” from probabilistic tyranny no doubt contributes to the more human-like appearance of the conversation with ChatGPT relative to other competitors and previous versions.

According to Wolfram, the neural network has about 175 billion “weights” assigned to it, and every time it generates a token it has to do calculations with all those weights (Wolfram is referring to GPT-3, the version prior to GPT-4 that ChatGPT incorporates at the time of writing this chapter). Thus, every time it adds a word (a token) to the text it generates, the system has had to do about 175,000 million numerical calculations (that is, apply the weights of each of the connections to the input numerical matrices to generate the output).

The acronym GPT stands for Generative Pre-trained Transformer and the last part is the newest technology. The so-called *transformer* (Vaswani et al. 2017) is the neural network architecture resource that allows the prediction of the following words to take into account those that have been processed before through attention heads (see chapter 5 in this volume). Since ChatGPT does not know what it is talking about, nor does it have any communicative intention, the programmers have found that the way to maintain some coherence is to “go back” and recalculate the probabilities of the next token.

Another central feature of ChatGPT is that it has been trained with feedback from human users, which largely explains its success in creating coherent text and in convincing users that they are conversing with it. Human readers identify instances of digression, repetition, and lack of coherence (or incivilities), which are then evaluated. Based on these evaluations, the system makes adjustments to its operations. Consequently, an additional auxiliary neural network is generated, which learns from these evaluations and modulates the output of the primary network.

An essential idea that we can already see is that ChatGPT does not operate with linguistic units, but with groups of characters—that is, letters. Therefore, in principle, it has no way (or ability) to link letter sequences with meanings. The only thing ChatGPT “knows” is a huge mathematical function that allows to calculate how often groups of characters appear next to other groups of characters, and it relies on that to generate “plausible” strings of characters, including the white spaces between them. Although it may sound derogatory, it deserves the nickname *stochastic parrot*. The only meaning that its answers can have is that provided by the users who read them (and which they inevitably tend to attribute to their inorganic interlocutor).

Many of the linguists and psychologists who claim that ChatGPT is a model of human language that can allow us to understand it better (and that can even refute some scientific approaches to it), seem to overlook that all the enormous training and computing effort that ChatGPT and its relatives imply (including enormous energy consumption) it is done precisely to be able to generate apparently coherent text *without understanding what is being said*. So, except from the nihilistic point of view that says that we don't understand what we say either (that we are Chinese rooms), it is difficult to imagine why anyone would

think that this program is a realistic model of what we humans do when we speak, given that we, although we do not know how, do understand, most of the time, what we say.

The majority of the remaining chapters in this volume examine this argument in depth by comparing the knowledge of language that humans possess and its use with the artificial knowledge of language that systems like ChatGPT possess. As discussed in some of these chapters, one of the issues arises from the fact that many researchers employ conceptualizations of language that are markedly distinct from those put forth by contemporary linguistics.

3. What would human language have to be like for ChatGPT to be a good model of it?

In fact, contrary to what I suggested at the end of the previous section, it is not so difficult to envisage why there are authors who do believe that ChatGPT is a model of human language: because they have an incomplete understanding of what human language is like. The typical line of reasoning would be as follows: since ChatGPT produces grammatical sentences and it only knows about probabilities of co-occurrence of words, then the grammar of languages is a question of probability of appearance of some words next to others.

To validate this logic, it is necessary to assume that ChatGPT generates grammatical sentences in a manner analogous to that of humans. Such assumptions are, however, difficult to justify without further argumentation. Although ChatGPT and its relatives are called language *models*, what they aim to model is not human language understood as a cognitive capacity (that is, the system of knowledge or faculty that allows us to learn and use any language), but the probability of appearance of groups of characters next to other groups of characters, based on the frequencies observed in the training texts. And that is not a model of human language (see especially chapter 3 by Lobina).

To better understand why it is not, we must pay attention to a crucial distinction in the history of linguistics as a science that many commentators tend to overlook: the contrast between an internalist perspective (language as a capacity) and an externalist perspective (language as a product).

Let's think for a moment about what English (or any other language) really is, where it exists, and what kind of object or entity that language (or any other) is. According to the internalist perspective on language (the perspective of linguistics as a cognitive science), English is a system of knowledge that has developed in the brains of people who speak English (for the sake of simplicity, think of so-called native speakers of this language). Let's call this system of knowledge, following Noam Chomsky (1986), the I-language (*i* for *internal* and *intensional*). An I-language minimally consists of a repertoire of concepts or meanings associated with chains of phonemes (words in the strict sense), and a set of grammatical rules to construct sentences with them. For its part, the E-language (*e* for *external* and *extensional*) would be the sum of all the sentences produced by the speakers of a given language, for example, the corpus of everything that has been written in English in the last three hundred years (to which we would still have to add all the sentences that the billions of speakers of that language in history have uttered without ever writing them, undoubtedly many more than those that have been written).

Internalist linguists believe that the object of study of linguistics as a cognitive science is the I-language (the system of knowledge in people's brains), while the E-language is an incomprehensible and incoherent object. Where English really exists is in the brains of its speakers, not "out there." Externalist linguists believe, on the contrary, that what really exists is the E-language, a kind of socio-cultural construct that is passed from generation to generation, with the I-language being an imperfect copy of the E-language in the brains of people developed in them by the practice of speech.

Regardless of who is right in this chicken-and-egg problem, what seems clear and indisputable is that ChatGPT is a model of E-language, not of I-language.

Why did modern linguistics shift from the study of the product of language to the study of the knowledge of language? Because the speakers of any language can create and understand hundreds of thousands of different sentences, which they have never heard before, and that fact requires an explanation.

It has been calculated that, on average, when we are talking we can choose about 10 different words after the last one we used. Sometimes only one is valid (like after "*helter _*"); on other occasions hundreds of them are possible (like after "*I would eat a _*"). Calculations by psycholinguist Steven Pinker (1999) indicate that the average is 10. If people can make sentences of 20 words (any of this text has quite a few more) and after each word speakers can choose from 10 other words, then the number of different sentences any speaker can create and understand is 10^{20} (that is, 100 trillion different sentences), and that drastically limiting the maximum length of a sentence to 20 words. Those 100 trillion twenty-word sentences alone would require a childhood of about 100 billion years to learn (assuming about five seconds to learn each sentence). Since this is obviously absurd, given that the human brain "only" has about 80,000 million neurons, internalist linguists assume that speakers have a generative system of rules, mechanisms or principles that allow them to create each of those 100 trillions of different twenty-word sentences that they could pronounce and understand if they lived long enough.

Does ChatGPT have that system? It cannot be said with certainty, but there is no evidence that this is the case (see chapters 5 and 7 for discussion). It is important to keep in mind that a central part of the generative mechanism of human language is the hierarchical structure of constituents, which basically implies that sentences in human language (English or any language) are not linear sequences of words (although they appear that way in speech or writing), but groupings of words in "boxes" or constituents, contained in each other, that determine the meaning that sentences have. The phrase *Deep blue sea* is ambiguous because the linear sequence of words can reflect both the hierarchical structure in which *deep* modifies *blue* (and the sea might not be deep) and the structure in which the adjective *deep* modifies *blue sea* (in which case the sea might be light blue). In the first case, we have the simplified structure of (1a), in which the sequence *deep blue* forms a constituent, and in the second, that of (1b), in which it does not:

- (1) a. [*Deep blue*] sea
- b. *Deep* [*blue sea*]

In (1a) *deep* cannot modify *sea* because it does not have structural access to that noun, while in (1b) *deep* has to modify *sea* and cannot access *blue*, although it lies at its side. There is no way to explain these asymmetries, ubiquitous in human language, by analyzing syntax as a linear sequence based on probabilistic chaining of words, which is what ChatGPT and other LLMs do. The syntactic structure is inaudible and invisible (and is

not present in the sound waves or in the linear sequences of words that form texts), but it is essential for interpreting the meaning of the sequences of words that reach us and for constructing the complex thoughts and propositions that humans create and (sometimes) communicate.

Already in 1957 Chomsky proposed the famous, absurd but perfectly grammatical sentence *Colorless green ideas sleep furiously*, precisely to show that the generation of human language cannot be modeled as a system of chaining the most probable (most expected) word with respect to the previous ones, but as a system that uses formal mechanisms independent of the meaning of each word. This design feature of language, that is, a generative mechanism that creates hierarchical structures using syntactic categories (noun, verb, preposition, etc.) independently of the lexical content of each item (*table*, *eat*, *of*, etc.) is what may actually help explain the ability of speakers of human languages to create and communicate new ideas and entertain thoughts free from stimulus control and independent of interactions with other people (see chapter 8 in this volume). It is important to note that such a syntax is necessary to create meanings (like the two in *Deep blue sea*), but not to chain words from left to right. ChatGPT does not use a hierarchical constituent structure simply because it does not need one, since it has nothing to say.

We have considered how LLMs work, and we have seen that there is no objective evidence of any type of grammatical or semantic structure in that task. If we return to the distinction between I-language and E-language, we are able to affirm that ChatGPT is a (mathematical) model of the E-language, not of the I-language. Therefore, it can be said that it is an efficient engineering tool, a notable achievement of AI, but not that it is a model of human language, that is, of the faculty that allows us to create new and original thoughts and translate them into sounds (and these in characters, like those handled by ChatGPT). So to speak, humans create character sequences as a result of previously combining concepts using a hierarchical recursive syntax, while ChatGPT creates character sequences from other character sequences. It is a stochastic parrot in a Chinese room.

Only those who truly believe that human languages are sets of flat texts (linear sequences of words/tokens) and believe that children learn to speak by memorizing and repeating the sentences they have heard, or by calculating the probability of the next word, without understanding its meaning, can continue to maintain that ChatGPT and its relatives are models of human language. To put it more clearly: ChatGPT is a mathematical model of the product of human language use, rather than a model of the capacity to produce human language.

It could be concluded, therefore, that stating that ChatGPT speaks (or writes) in English (or in any other language in which it has been trained) is the same as equating the procedure Diego de Velázquez used with the brush to paint *Las Meninas* with that of the photographer who reproduces the famous painting. The result may be superficially similar, but the process of creation is radically different, and we will not learn much about Velázquez's creative technique by analyzing the process of developing a photograph of *Las Meninas*.

4. Has ChatGPT refuted Chomsky (or any other linguist)?

Not all scientists who believe that ChatGPT is a model of human language (even capable of deciding which theories of language are most appropriate) are “flat syntaxers” (in the sense that they deny the existence of the hierarchical structure of constituents). Some, despite acknowledging that human language does not consist of mere strings of words, posit that ChatGPT must have, *in some manner*, discerned the underlying principles that enable the creation of these structures from the training corpus. Thus, while assuming that ChatGPT is a good model of the human mind (although those who designed it did not intend to do so), they affirm that it has disproved the hypothesis (usually associated with Chomsky) that there is an innate bias for language in humans. The reasoning goes, more or less, like this: since ChatGPT has not been equipped with the principles and mechanisms that some linguists assume must be innate in order to explain language learning (for example, the generative mechanisms for producing hierarchical syntactic structures recursively), then, *since ChatGPT has learned those principles*, the nativist approach is false.

Cognitive psychologist Seven Piantadosi is quite clear about this:

“The rise and success of large language models undermines virtually every strong claim for the innateness of language that has been proposed by generative linguistics. Modern machine learning has subverted and bypassed the entire theoretical framework of Chomsky’s approach, including its core claims to particular insights, principles, structures, and processes” (Piantadosi 2023: 1).

It is noteworthy that the argument employed by Piantadosi to refute the notion of these systems being mere repetitive parrots is identical to the one previously utilized by Chomsky in the 1950s and 1960s to substantiate the assertion that if humans acquired the capacity to speak through mere imitation or replication of environmental speech, their ability to construct novel sentences would remain unexplained:

“If models only repeated what they had seen before, they would not be able to generate anything new, particularly complex sentence structures that are grammatical and coherent” (Piantadosi 2023: 5).³

Of course, although it would be surprising, the nativist approach could be false. But ChatGPT does not present any evidence in that direction, simply because it is not possible to assert that it has learned any language. No matter how sophisticated its responses may seem, ChatGPT does not understand a word of what it itself writes. ChatGPT has not internalized any grammar with which to create and combine concepts, nor does it have concepts, nor, therefore, has it acquired any language through learning, as human beings do. As I suggested before, for ChatGPT to be a model of human language, people would have to use language without understanding it, using linear sequences more likely in certain contexts, rather than constructing new thoughts using an internal grammar and then converting them into strings of words for communication.

Anticipating an issue that is discussed in more detail in later chapters, let us consider an example of how we could know that ChatGPT has not genuinely acquired a human language and, therefore, has not been able to deduce the rules of syntax from the

³ And more than curious, it is ironic that a computational model called Generative Pre-Trained Transformer (GPT) is supposed to be the one that would have demonstrated that so-called Generative Transformational Grammar is an illegitimate scientific enterprise.

statistics of words. Computational linguist Roni Katzir (2023) asks ChatGPT which of the following two sentences is more acceptable:

- (2) a. *The person that Mary met yesterday and that John will talk to Ed about tomorrow arrived.*
b. *The person that Mary met yesterday and that John will talk to Ed about his new idea tomorrow arrived.*

Although both are complex sentences, any native English speaker who is 9 or 10 years old is aware that the first is a grammatical sentence and the second is not. This is despite ChatGPT indicating that the second sentence is more acceptable than the first (to the extent that such a question is relevant).

In essence, what happens in this type of construction is that when two relative clauses are coordinated (in this case that modify *the person*), the “gap” left by the relative pronoun has to be the same in both. Thus, we see that in (2a) the object of *met* is missing and the object of *talk about* is missing too, while in (2b) the object of *met* is missing, but the object of *talk about* is not missing (*his new idea* has been added that fill that gap). Of course, that creates an ungrammatical sentence (there are two conflicting complements of *talk about*), although ChatGPT prefers it, simply because, although it doesn’t understand what it means, it knows that *his new idea* is very likely to appear after *talk about*.

It should be noted that the restriction that appears in the example of (2a), that is, if there is a “gap” in one coordinate there must also be one in the other, is a formal restriction to the syntax of human languages already described in 1960s by generativist linguists, a restriction that human children acquire effortlessly and with much less training data than ChatGPT.

The conclusion, therefore, is twofold: ChatGPT has not internalized the mechanisms and principles that humans use to construct sentences and, therefore, in no way can it be said that this LLM has demonstrated that the proposals of generative grammar (or any theory of language acquisition) are false. Simply put, the inductive biases that humans present and those presented by LLMs such as ChatGPT are totally different, so it makes no sense to consider the latter a model of the former (see Chapters 2, 3 and 4 for different approaches to this issue).

As Katzir concludes in his informative response to Piantadosi, “Using such models to write entertaining poems and short stories is one thing. Using them to understand the human faculty of language instead of doing actual linguistics is quite another” (Katzir 2023, 9).

5. Therefore, linguists have nothing to learn from LLMs?

Surely, that would be a hasty conclusion. The human faculty of language is a very complex system that actually includes different capacities that interact with each other. The traditional areas of language study (semantics, pragmatics, syntax, morphology, phonology, or phonetics) designate different systems of knowledge, which, when integrated, give rise to what is commonly referred to as the Faculty of Language (FL). Consequently, it is conceivable that LLMs may more accurately reflect the characteristics of some of these systems while being less pertinent to others.

As I have noted before, the divergence in perspectives regarding the degree to which natural and artificial linguistic knowledge are analogous is closely associated with the broader discrepancy between internalist and externalist approaches to language and languages. As we have seen, for the Chomskyan point of view, a language is a state of the language faculty, that is, it is a system of knowledge, a cognitive organ, while for the opposite point of view, a language is an external system, a cultural object that the brain is somehow able to assimilate and represent. This discrepancy underlies the different assessment that both traditions make about the relative weight that biology and culture should have in the explanation of the nature of language. But, in reality, it is a false problem, since language knowledge involves both biological and cultural aspects. As I have argued elsewhere (Mendívil-Giró 2019) the aforementioned traditions, rather than being in contradiction, could be considered complementary if we take into account the specific aspects of language to which they refer. If this is the case, it is possible that the discrepancy in the assessment of the extent to which ChatGPT's knowledge of language resembles that of a human being could also be understood as a misunderstanding about the type of human linguistic abilities that are being considered.

For greater concreteness, let us consider the influential model of the FL proposed by Hauser, Chomsky and Fitch (2002), according to which FL could be conceived as a complex system minimally integrated by three independent components: (i) a conceptual-intentional (CI) system, related to meaning and interpretation; (ii) a sensorimotor system (SM), related to the perception and production of linguistic signals, and (iii) a computational system (CS), syntax in the strict sense, responsible for the creation of the recursive and productive syntactic structure that underlies linguistic expressions.

In subsequent work, Chomsky (2007) proposed that the relationship between the computational system and the conceptual-intentional (CI) and sensorimotor (SM) systems is asymmetric. Specifically, he suggested that the computational system (syntax) evolved in response to the CI system, forming an internal language of thought (ILOT) that is used primarily for representing reality and creating thought: "the earliest stage of language would have been just that: a language of thought, used internally" (Chomsky 2007, 13). This ILOT, which is essentially common to the species, would subsequently have been connected to the SM system for externalization and, therefore, for communication. In accordance with this hypothesis, externalization would be a process susceptible to fluctuations in the environment, thereby rendering it prone to change and diversification.

As argued in Mendívil-Giró (2019), the Hauser, Chomsky and Fitch model must be capable of representing the diversity of languages. People do not speak "language", but rather specific languages such as English, Russian and Chinese. Therefore, the FL (the I-language of each individual) must also include a component derived from the environment (that is, internalized), whose function would be to systematically connect the syntactic-semantic derivations generated by the ILOT (resulting from the interaction between the conceptual and computational systems) with the sensorimotor system. The crucial idea now is that this component, which is learned, is what really differentiates languages from each other and constitutes the genuinely cultural component of every human language.

For the sake of expository convenience, we shall refer to this component as the *externalization interface*. Such an externalization interface can be conceived of as an area of long-term memory that provides a stable connection between, on the one hand, purely conceptual (semantic) elements as constructed by the computational system (syntax) and,

on the other hand, the sensorimotor systems that process and produce the material linguistic signals that human beings perceive and produce when they use language for communication. This interface, which may be conceived of as a collection of formants or exponents, can be identified with traditional morphology and phonology.

Three of the components of FL—the conceptual-intentional system, the computational system, and the sensorimotor system—are essentially universal, as they have a robust biological conditioning. In contrast, the fourth component, the externalization interface, although the systems it interacts with strictly condition it, is culturally variable, as it is the result of learning through environmental stimuli. From this perspective, language acquisition does not entail the acquisition of the entire system of knowledge (the I-language), but rather the learning of one of its components (the externalization interface). The development of language in the individual can be considered to be equivalent to the growth of this interface in the mind/brain. This can be expressed in other words as the process of learning to externalize the ILOT in a manner that is consistent with the ways in which members of our community do so.

It is important to note that ChatGPT (like the majority of LLMs) is not designed to represent reality or create thoughts. Therefore, it does not require a conceptual-intentional system or a syntax. In fact, ChatGPT appears to operate solely with elements of the externalization component, specifically a surface morphology (tokens) and a secondary version of phonology (writing).

This schematic model of a person's FL provides a useful framework for understanding the divergence of views in contemporary linguistics regarding the impact of culture and biology on language design, and, in turn, on the language learning capacity of LLMs. The crux of the matter is that when linguists are in disagreement on these issues, it is highly probable that they are in fact discussing disparate concepts, not only with regard to the term *model*, but also with respect to the term *language*.

A significant point of divergence in interpretation arises from the insufficient differentiation within the externalist tradition between the entire I-language and one of its constituent elements, namely the externalization interface. Statements such as that 'languages are external to the brain', that 'they can vary without limits', or that 'they are learned using general statistical learning resources', can be more adequate if they refer to the repertoire of formants for externalization, than if they refer to the entire set, to the complete I-language. It seems reasonable to posit that the externalization interface is, to a large extent, a cultural construct. However, it is not reasonable to conclude that it is a language in the strict sense.

Indeed, there is some evidence that the developmental patterns of syntax and of the externalization interface are distinct. This evidence can be classified into two categories, pertaining to the asymmetry in acquisition and the asymmetry in the degree of interlinguistic variation of each component. Firstly, the poverty of stimulus argument is particularly compelling when applied to the development of syntactic knowledge, as opposed to morphological and phonological knowledge (see Lidz and Gagliardi 2015). Secondly, it is more straightforward to make descriptive generalizations about syntax than about morphology. This is due to the fact that syntactic structures (phrases, sentences) and syntactic relations (subject of, object of, argument, modifier, etc.) are considerably more general in languages than morphological categories (such as root, suffix, gender, or inflection). It is therefore possible that the linguistic knowledge developed by LLMs based

on the statistics of formant co-occurrence is more similar to human knowledge of morphology or phonology than to human knowledge of syntax and semantics (for further discussion of this, see chapter 6 in this volume, with reference to the case of morphology). In light of the aforementioned considerations, it can be posited that the rejection of the notion that LLMs are models of the human faculty of language (as previously stated in this chapter and elaborated upon in other chapters of this volume) does not necessarily preclude the relevance of such models for linguistic research, particularly given the intricate nature of language as a system subject to different factors influencing its development.

6. What the reader will find in the following chapters

In Chapter 2 (*The quo vadis of the relationship between language and large language models*), Evelina Leivada, Vittoria Dentella and Elliot Murphy undertake a comprehensive examination of the pivotal issues that underpin the motivation behind this volume. These include the extent to which LLMs can be considered models of human language and the extent to which they can contribute to our understanding of the processes by which humans acquire and use languages. The authors undertake a critical analysis of certain proposals which advocate the adoption of LLMs as scientific models of language. They highlight that the utilization of non-transparent theoretical scientific models carries with it a number of theoretical and empirical risks. The authors approach the question of what constitutes a scientific model from four complementary angles: the object (the target system being represented), the medium (the system doing the representing), the meaning (the content of the representation), and the user (the agent using the representation). Despite concluding that LLMs currently offer limited insights into human language, Leivada, Dentella, and Murphy conclude their chapter with an interesting outlook on potential future research directions that could enhance the capacity of AI development to effectively aid in the study of human language knowledge.

Following the same line of argument, David Lobina (in Chapter 3 *Language models as function approximators of text data: disrupting comprehension through an adversarial attack*) delves into the internal functioning of LLMs and shows that, as examples of Machine (Deep) Learning models, they are best regarded as “function approximators”, or vast correlation machines. Given a dataset of inputs and outputs, a Machine Learning network will construct a function that best describes (approximates) the distribution in the data, and from whence it can generate similar input-output pairs if prompted. In the case of ChatGPT, the dataset was composed of text—the products, or externalizations, of the language faculty. In this sense, Lobina concludes that ChatGPT does not model a natural language per se, but the statistical distribution of text. In his view, the claim that some of the results emanating from the development of LLMs bear any consequence for the study of language, especially regarding linguistic competence and acquisitional studies, appear to be the case of committing a category mistake.

In chapter 4 (*On modern language models, impossible languages, and anti-Science*), Dan Milway makes a similar point, warning of the dangers of using systems developed as engineering products as if they were explanatory theories. Paying particular attention to the problem of impossible languages, Milway critically reviews Piantadosi’s (2023) arguments for his claim that LLMs are refutations of rationalist theories of language, along with his responses to criticisms of these arguments. The author finds that Piantadosi

misrepresents the claims, predictions, and arguments of contemporary rationalist theories of language. He posits that the objectives of rationalist theories of language and LLMs are fundamentally disparate: the former is a scientific inquiry, driven by the pursuit of explanation and understanding, whereas the latter is an engineering project, focused on the development of tools. Confounding these two distinct approaches can result in an anti-scientific stance being inadvertently adopted.

The next three chapters focus on the investigation of the knowledge of language developed by LLMs and in the methods to discover it.

Chapter 5 (*What kind of linguistic knowledge is encoded in Large Language Models?*), by Lucía Pitarch and Miguel López-Otal, focuses on understanding how LLMs represent aspects of grammar and semantics. Examining models such as BERT and ChatGPT, the discussion delves into the methods used to explore linguistic knowledge within LLMs. The first section examines research claiming that, despite being trained on unstructured text, these models internally acquire human-like grammatical knowledge. The second part of the chapter focuses on the exploration of semantic knowledge, addressing the uncertainties surrounding how language models learn meaning and semantics. Lexico-semantic relations and figurative language are examined in monolingual and multilingual contexts to assess the encoding of semantic information in LLMs.

In chapter 6 (*ChatGPT and linguistic theory, with a focus on morphology*), Stela Manova proposes to move beyond the abstract level at which the discussion of the extent to which LLMs can serve as theories of language has been conducted. Thus, she looks at how LLMs work technically and what their internal organization means for linguistic theory. Since algorithms in different LLMs may differ, Manova focuses on ChatGPT, paying particular attention to the so-called Byte Pair Encoding (BPE), a form-based subword tokenization algorithm that is at the core of ChatGPT. The author illustrates how this algorithm treats language and compares it to current research in theoretical morphology. Following the logic of the Big O notation in computer science, she argues that a form-based analysis is less complex than a semantics-based one, suggesting that ChatGPT could be a suitable model of human knowledge of morphology.

Chapter 7 (*Evaluating neural language models as cognitive models of language acquisition*) continues the investigation of the syntactic knowledge of LLMs within the context of language acquisition theory. Héctor Javier Vázquez Martínez, Annika Heuser, Charles Yang and Jordan Kodner argue that some of the most prominent benchmarks for evaluating the syntactic capacities of LLMs may lack sufficient rigor. In particular, the authors demonstrate that the template-based benchmarks lack the structural diversity that is commonly observed in theoretical and psychological studies of language. When trained on small-scale data modeling child language acquisition, language models can be readily matched by simple baseline models. They recommend the use of readily available, carefully curated datasets that have been evaluated for gradient acceptability by large pools of native speakers and are designed to probe the structural basis of grammar specifically. In examining the LI-Adger dataset, the authors demonstrate that language models evaluate sentences in a manner that is inconsistent with the linguistic behavior observed in human language users. In light of these findings, the authors conclude with recommendations for more effectively integrating language models with empirical studies of child language acquisition.

Vincent Carchidi's final chapter (*The creative aspect of human language use: how modern language models fare with an old idea*) turns from the problem of the nature of artificial knowledge of language and its development to the problem of language use. He considers to what extent the success of LLMs in their ability to produce human-like text sheds any light on the creative aspect of language use (CALU), that is, the notable fact that human language use is stimulus-free, unbounded, yet appropriate and coherent to the circumstances of its use. According to Carchidi's analysis, LLMs fail to reproduce this ability without exception. Moreover, he shows that they offer no account of how humans acquire the ability to use their language and what role this should play in a theory of human language. He argues that Piantadosi's (2023) critique of Chomsky's approach to linguistics is misguided, concluding that the failure to shed any scientific light on CALU amounts to a deficiency of a methodology that considers LLMs accurate models of human language, and that this fact is a vindication of the so-called 'Galilean' style of inquiry employed by Chomsky.

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